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Electrochemotherapy in radiotherapy-resistant epidural spinal cord compression in metastatic cancer patients

Authors:

Frederic Deschamps[1,2], M.D. ; Ph. D.

Lambros Tselikas[1], M.D. ; Ph. D.

Steven Yevich[1], M.D.

Baptiste Bonnet[1], M.D.

Charles Roux[1], M.D.

Adrian Kobe[1], M.D.

Benjamin Besse[3], M.D. ; Ph. D.

Kevin Berthelot[4], M.D.

Amelie Gaudin[5], Pharm.D.

Lluis M. Mir[2], Doctorat Honoris Causa

Thierry de Baere[1], M.D. ; Ph. D.

Affiliations:

1: Department of Interventional Radiology, Gustave Roussy, Université Paris-Saclay, 114 rue Edouard Vaillant, 94805, Villejuif, France.

2: Université Paris-Saclay, CNRS, Gustave Roussy, Aspects métaboliques et systémiques de l'oncogénèse pour de nouvelles approches thérapeutiques, 94805, Villejuif, France.

3: Department of Medical Oncology, International Center for Thoracic Cancers (CICT), Université Paris-Saclay, Gustave Roussy, 114 rue Edouard Vaillant, 94805, Villejuif, France.

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4: Department of Radiotherapy, Gustave Roussy, Université Paris-Saclay, 114 rue Edouard Vaillant, 94805, Villejuif, France.

5: Department of Pharmacology, Gustave Roussy, Université Paris-Saclay, 114 rue Edouard Vaillant, 94805, Villejuif, France.

Corresponding author: Dr. Frederic Deschamps

Department of Interventional Radiology, Gustave Roussy, Université Paris-Saclay, 114 rue Edouard Vaillant, 94805, Villejuif, France.

Frederic.deschamps@gustaveroussy.fr

00 33 1 42 11 42 11

ABSTRACT

Objective: To report efficacy and safety of percutaneous electrochemotherapy (ECT) in patients with radiotherapy-resistant metastatic epidural spinal cord compression (MESCC).

Material/ methods: This retrospective study analyzed all consecutive patients treated with bleomycin-based ECT between February-2020 and September-2022 in a single tertiary referral cancer center. Changes in pain were evaluated with the Numerical Rating Score (NRS), in neurological deficit with the Neurological Deficit Scale, and changes in epidural spinal cord compression were evaluated with the epidural spinal cord compression scale (ESCCS) using an MRI.

Results: Forty consecutive solid tumour patients with previously-radiated MESCC and no effective systemic treatment options were eligible. With a median follow-up of 5.1 months [1-19.1], toxicities were temporary acute radicular pain (25%), prolonged radicular hypoesthesia (10%), and paraplegia (7.5%). At 1-month, pain was significantly improved over baseline (median NRS: 1.0 [0-8] versus 7.0 [1.0-10], *P* <.001) and neurological benefits were considered as marked (28%), moderate (28%), stable (38%), or worse (8%). Three-month follow-up (21 patients) confirmed improved over baseline (median NRS: 2.0 [0-8] versus 6.0 [1.0-10], *P* <.001) and neurological benefits were considered as marked (38%), moderate (19%), stable (33.5%), and worse (9.5%). One-month post-treatment MRI (35 patients) demonstrated complete response in 46% of patients by ESCCS, partial response in 31%, stable disease in 23%, and no patients with progressive disease. Three-month post-treatment MRI (21 patients)

demonstrated complete response in 28.5%, partial response in 38%, stable disease in 24%, and progressive disease in 9.5%.

Conclusion: This study provides the first evidence that ECT can rescue radiotherapy-resistant MESCC.

Keywords: Electrochemotherapy, Therapy, Palliative, Bone, Metastasis, spinal column, Extramedullary Spinal Cord Compression, Paralysis, legs, Radiology, Interventional

INTRODUCTION

Metastatic epidural spinal cord compression (MESCC) frequently results in severe pain and neurologic impairment such as paraplegia. Conventional external beam radiation therapy is the standard of care for frontline therapy[1]. Unfortunately, recurrence is common with local failure rates ranging from 30% to 86% at 2-years, with this high variability explained by differences in tumour radiosensitivity, MESCC size, and disease extension[2]. In patients with MESCC recurrence after radiotherapy, therapeutic approaches pose major challenges. The option of re-irradiation is often limited out of concern for spinal cord damage from cumulative radiation dose. Systemic treatment may be effective however the time to response is usually too long for lesions with rapid neurologic progression, leaving the patient with permanent neurological damage. New local treatment modalities are needed to address this therapeutic dead-end.

Electroporation is a minimally-invasive therapy that has been explored in challenging tumour locations[3-5]. Needle-like electrodes are positioned within or straddling a tumour lesion, and high-intensity electric pulses are generated across the tumour tissue causing cellular membrane damage. The degree of damage depends upon the electric pulse amplitude and number of applied pulses. Exceeding a threshold of 400 Volts/cm can cause temporary damage, termed reversible electroporation. Exceeding a threshold of 800-1000 Volts/cm can result in permanent loss of cell homeostasis and cell death, termed irreversible electroporation[6]. Hence, while irreversible electroporation carries a risk of possible permanent spinal

cord injury, the temporary damage caused by reversible electroporation may provide a novel and unique treatment opportunity.

Electrochemotherapy (ECT) refers to the concomitant application of reversible electroporation and intravenous administration of chemotherapy, with ECT enhancing chemotherapy uptake into the tumour cells. Bleomycin's mechanism of action induces mitotic death, which is theoretically safe, or at least less toxic, for non-mitotic cells such as neurons. When reversible electroporation causes temporary cell wall damage, the injected bleomycin can enter the tumour cell cytoplasm, and generate DNA double-strand breaks that cause arrest in the G_2-M phase of the cell cycle (radiomimetic effect)[7]. In addition, ECT has a vascular disrupting action, known as the "vascular lock" effect, via destruction of small tumour blood vessels[8].

Published literature on ECT in the context of bone tumours is relatively nascent, but early findings support the potential of this approach in MESCC. A few clinical trials have demonstrated safety in the treatment of painful bone metastases located in the appendicular skeleton[9-10]. In the treatment of lumbar spine metastases, a few case reports demonstrated a substantial pain palliative effect without nerve root damage[11-12]. A numerical modeling study that considered the hypothetical treatment of 3 thoracolumbar vertebral metastases suggested nearly 100% tumour treatment, but with a hypothetical potential for permanent spinal cord damage as the electric field required was above the assumed threshold for irreversible electroporation (>800 V/cm)[13]. Although the effect of electroporation on spinal cord tissue has not been thoroughly investigated,

recent preclinical studies have suggested that there may be neural regeneration even after irreversible electroporation[14-15].

This study reports our initial experience of ECT for MESCC in terms of safety, clinical efficacy (measured in terms of pain and neurological symptoms), and local tumour control. To our knowledge, no clinical data have been published on this technique for MESCC.

MATERIALS AND METHODS

A retrospective study was performed by reviewing medical records for all patients treated with ECT for MESCC between February 2020 and September 2022 in our tertiary referral cancer hospital. To be eligible for this study, patients had to have a metastatic solid tumour, have undergone prior radiotherapy for MESCC, have absence of or contraindication for re-irradiation or rapidly effective systemic treatment options, and have undergone ECT for MESCC. Patients treated with MESCC below the conus medullaris were excluded. The decision to proceed with ECT was systematically made by a multidisciplinary tumour board.

Electrochemotherapy

ECT was performed by two interventional radiologists (FD; CR), each with more than 5 years of experience in performing therapeutic bone procedures. For all procedures, patients were intubated and positioned prone within a surgical suite equipped with both fluoroscopic and computed tomography (CT) imaging (INFINIX-i 4D-CT^{TM}, Canon Medical Systems, Otawara, Japan). Several commercially available 17G electrodes were percutaneously inserted (IGOE710, IGEA, S.p.A, Via Parmenide 10/A, 41012 Carpi MO, Italy) though a bone access needle and distances between electrodes were measured with CT. Electrode quantity and configuration were dependent upon MESCC location and extent to create an electric field that encompassed the entire tumour volume. Once inserted, all electrodes were connected to a pulse generator (Cliniporator Vitae, IGEA, S.p.A, Via Parmenide 10/A, 41012 Carpi MO, Italy). Bleomycin (15 mg/m^{2,} Bleomycin Sanofi, France) was administered intravenously in a

single infusion over 1 minute. Eight minutes after the completion of the infusion, reversible electroporation was performed with two series of four electric pulses of opposite polarity, of 100 µs duration and 1 kHz pulse repetition frequency. Reversible electroporation impacts were only delivered between electrodes located less than 5 cm apart. As recommended by the manufacturer, we applied a voltage-to-distance ratio of 1000 V/cm for each pair of electrodes except for pairs located on each side of the central canal. In this electrode configuration, we decreased the voltage-to-distance ratio to 600 V/cm to minimize the risk of definitive electrical damage to the spinal cord.

Delivery of the electrical pulses was synchronized with the ECG and triggered during the refractory phase of the heart. A rocuronium dosage was administered to obtain deep muscle blockage using a train of four watch (TOF) device. Upon patient awakening, the effect of curare was resolved by administering sugammadex (24 mg/kg according to the degree of neuromuscular block).

After ECT completion, the electrodes were removed and a final CT was performed to evaluate for eventual immediate post-procedure complications, such as bleeding. The total duration of the ECT procedure was recorded as the time between the initial CT, performed at the very beginning of the intervention to plan the electrode trajectories, and the final post-procedure CT.

Outcomes analysis

Safety was assessed based on treatment-related complications

during ECT follow-up. Complications were graded using Common Terminology Criteria for Adverse Events (CTCAE) v5.0[16].

Pain was evaluated at pretreatment, 1-month and 3-months using self-reported pain intensity on a scale of 0 to 10 according to the Numeric Rating Scale (NRS). Changes in pain versus pretreatment were categorized as marked improvement (decrease ≥50%), moderate improvement (decrease between ≥30% and <50%), stable (+/-30%), and worse (increase ≥30%)[17]. In addition, the total morphine milligram equivalents (MME) per day (in milligrams) per day before the ECT were compared to 1 and 3 months after treatment.

Neurological symptoms were evaluated using the Neurological Deficit Scale (NDS)[18], which is based on clinical examination, and ranges from grade A (no symptoms) to E (complete paralysis) (Table 1).

Grade A	No symptom
Grade B	Focal minor symptom such as pain
Grade C	Functional paresis with muscle strength 4/5, where the involved muscle is functional and ambulatory
Grade D	Nonfunctional paresis with muscle strength ≤3/5, where the involved muscle is nonfunctional or non-ambulatory
Grade E	Complete paralysis or urinary and rectal incontinence.

Table 1: *Neurological Deficit Scale (NDS).*

Neurological symptoms were assessed at pretreatment, 1-month and 3-months and benefit was categorized as marked neurological improvement versus pretreatment (decrease ≥2 grades), moderate (decrease of 1 grade), stable (same grade), and worse neurological deficit (increase ≥1 grade).

Epidural tumour involvement was evaluated using the epidural spinal cord compression scale (ESCCS)[18]. The ESCCS is based on magnetic resonance imaging (MRI) and ranges from grade 0 (no epidural involvement) to 5 (complete block) (Table 2).

Grade 0	Vertebral bone involvement only without canal compromise
Grade 1	Minimal involvement of the epidural fat but without thecal sac displacement
Grade 2	Impingement and/or displacement of the thecal sac
Grade 3	Impingement of the spinal cord, but without significant distortion or displacement of the spinal cord
Grade 4	Spinal cord compression and displacement with cerebrospinal fluid still visible within the thecal sac

Table 2: Epidural spinal cord compression scale (ESCCS).

Local tumour control was assessed at 1-month and 3-months. In the absence of an established score to describe modifications to epidural extension after local treatment, we categorized our data as complete response (CR) when epidural extension had entirely disappeared (grade 0), partial response (PR) when epidural extension remained visible but had decreased at least 1 grade compared to pretreatment, stable disease when

ESCCS grade remained unchanged, and progressive disease when the volume of epidural extension had increased at least 1 grade on ESCCS.

Statistical Analyses

Patient characteristics were summarized using descriptive statistics (median [range] for continuous variables and proportions and frequencies for categorical measures). T-tests were used for comparisons. Analyses were performed using R software version 4.2.2. $P<.05$ was considered to be statistically significant. Fisher exact test were used to evaluate the correlation between the overall tumour response at 1 month (complete and partial) and the number of levels involved (1 or 2 levels versus 3 or more) and also to evaluate the correlation between the overall tumour response at 1 month and the vascularity of MESCC at pre-treatment MRI (hypervascular versus hypovascular).

RESULTS

A total of 40 patients with MESCC were included in this analysis (Table 3). All patients had MESCC at the spinal cord level. Median age was 58.4 [26-75] years, and the most frequent primary tumours were thyroid cancer (n=6), paraganglioma/ adrenocortical carcinoma (n=6), and lung cancer (n=5). Thirty-one patients had a neurologic deficit, 40 patients had pain with a median NRS pain score of 7.0 [1.0-10]. All patients has a radiotherapy, mean time to previous radiotherapy was 13 [1.5-30] months. Twelve patients had a previous local surgery.

	N= 40
Age in years, median [range]	58.4 [26-75]
Sex, male/female	23 /17
American Society of Anesthesiologists (ASA): score 1 /2 /3	1 /19 /20
Ongoing systemic chemotherapy	32 (80%)
Primary tumour	
Thyroid	6 (15%)
Paraganglioma/ adrenocortical carcinoma	6 (15%)
Lung	5 (13%)
Head and neck	4
Sarcoma	3
Kidney	3
Thymoma	2
Colorectal	2
Pancreas	2
adenoma carcinoma of unknown primary	2
Other[a]	5
Neurological deficit scale, grade A /B /C /D /E	1 /8 /14 /8 /9
Numeric Rating Scale (NRS), median [range]	7.0 [1.0-10]
Total opioïd doses in morphine equivalents daily (in grams)	**151.7 +/- 124.8 [0-400]**
Vertebral metastases with epidural extension (MESCC)	
Location	
Between C5 and T4	*10*
Between T5 and T9	*14*
Between T10 and L1	*16*
Hypervascular/ hypovascular	36/4
No. contiguous levels, 1 /2 /3 /4 /5	21 /9 /5 /3 /2
ESCCS grade 1/2/3/4/5	0 /9 /10 /9 /12

Previous local treatments at same level	
cEBRT /SBRT	*34 /6*
Thermal ablation /vertebroplasty /thermal ablation and vertebroplasty	*2 /10 /4*
Laminectomy / laminectomy + osteosynthesis	*5 /7*

Table 3: *Patient and vertebral metastases characteristics prior to undergoing electrochemotherapy.*

cEBRT: conventional external beam radiation therapy; ESCCS: Epidural Spinal Cord Compression Scale; SBRT: stereotactic body radiation therapy

a. neuroendocrine (n=1), melanoma (n=1), breast (n=1), prostate (n=1), chordoma (n=1)

Median procedure time for ECT was 90 [50-140] minutes and post-treatment hospitalization duration was 1 [1-5] day. Complications associated with the procedure were acute radicular pain that resolved within a few days in 10 patients (25%), prolonged radicular hypoesthesia in 4 patients (10%), sub-acute paraplegia in 2 patients (5%), and acute paraplegia in 1 patient (2.5%). Acute radicular pain and prolonged radicular hypoesthesia were considered mild (grade 1). Radicular pain typically occurred within 48 hours after ECT and disappeared within 4 weeks. Radicular hypoesthesia presented immediately after ECT and persisted for at least 3 months. Paraplegia was reported in three patients (7.5%) and considered severe (grade 3). In one patient, the paraplegia occurred immediately (acute) after the procedure while the deficit occurred after a delay of 2 weeks in two patients (sub-acute). MRIs demonstrated complete tumour response associated with intramedullary high signal intensity on T2-weighted image in all three cases. Despite steroid therapy, paraplegia had not improved at the 3-month follow-up visit.

The median follow-up period was 5.1 [1-19.1] months. All patients were seen at 1-month post-ECT, and 21 patients (53%) at 3-months; 15 patients had died due to progressive disease and 4 patients were lost to follow-up (Table 4).

Follow-up **(n patients)**	1 month (n=40)	3 months (n=21)
Median NRS	7.0 [1.0-10] vs. 1.0 [0-8]*	6.0 [1.0-10] vs. 2.0 [0-8]*
Pain improvement:		
- Marked (NRS decrease ≥50%)	60% (n=24)	62% (n=13)
- moderate (NRS decrease >30%)	20% (n=8)	5% (n=1)
- Stable	20% (n=8)	33.5% (n=7)
- Worse (NRS increase >30%)	0%	0%
Median MME per day (mg)	122 +/-125 vs. 151.7 +/- 124.8	75 +/-79 vs. 128 +/-134
Neurological benefit:		
- Marked (NDS decrease ≥2 grades)	27.5% (n=11)	38% (=8)
- moderate (NDS increase 1 grade)	27.5% (n=11)	19% (n=4)
- Stable	37.5% (n=15)	33.5% (n=7)
- Worse (NDS increase ≥1 grade)	7.5% (n=3)	9.5% (n=2)
Response at MRI	(35 patients)	(21 patients)
- Complete (ESCCS grade 0)	46% (n=16)	28.5% (=8)
- Partial (ESCCS decrease ≥1 grade)	31% (n=11)	38% (n=6)
- Stable	23% (n=8)	24% (n=5)
- Progression (ESCCS increase ≥1 grade)	0%	9.5% (n=2)

Table 4 : *Clinical and radiological outcomes.*

*NRS : Numerical Rating Score. MME : morphine milligram equivalents. NDS : Neurological deficit scale. ESCCS : Epidural spinal cord compression scale. * means statisticaly significant (p<0.001).*

The median NRS significantly decreased from 7.0 [1.0-10] before ECT to 1.0 [0-8] at 1-month (P <.001). For the 21 patients evaluable at 3-months, median NRS decreased from 6.0 [1.0-10.0] to 2.0 [0-8], (P <.001). At 1-month, 80% of patients reported an improvement in pain (marked in 60% of patients and moderate in 20%) and 67% at 3-months (marked in 62% and moderate in 5%). Moreover, 55% of patients reported neurological benefits at 1-month (marked in 28% and moderate in 28%) and 57% of patients at 3-months (marked in 38% and moderate in 19%).

Thirty-five patients (88%) had a post-treatment MRI at 1-month and 21 patients (53%) at 3-months. MRI follow-up demonstrated a tumour response in 77% of patients at 1-month (CR in 16 patients, PR in 11 patients) (Figures 1 and 2). Responses persisted at 3-months in 67% of the 21 patients evaluated (CR in 6 patients, PR in 8 patients). The overall tumour response at 1 month was significantly higher when MESCC involved 1 or 2 levels (87%, n=26/30) than when MESCC involved 3 levels or more (10%, n=1/10), p<0.005. The overall tumour response at 1 month was also significantly better for hypervascular disease (75%, n=27/36) when compared to hypovascular tumour (0%, n=0/4), p<0.005.

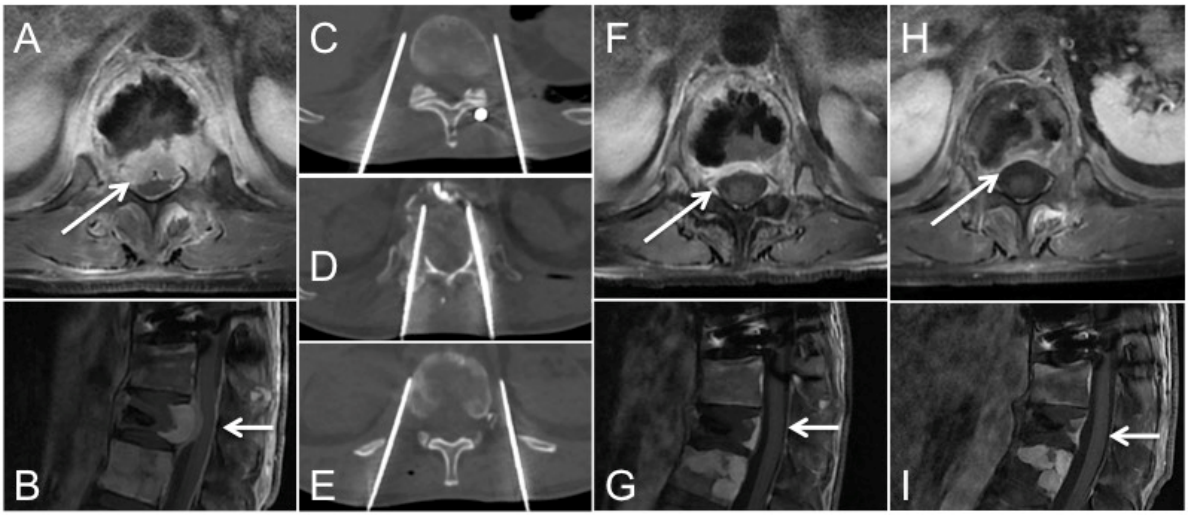


Figure 1 : *A 39-year-old male with metastatic paraganglioma and painful spinal cord compression at the T12 level (white arrow). The patient had undergone previous radiotherapy (20 Gy in 5 fractions) and cementoplasty at the same level, and had no ongoing chemotherapy. The pretreatment MRI (A and B) demonstrated a single level epidural spinal cord compression (grade 4). Electrochemotherapy was performed using 6 electrodes (C, D, E). The patient had marked pain improvement (pain decrease from 7/10 at baseline to 2/10 at 1-month and 0/10 at 3-months) and a marked neurologic benefit (neurological deficit scale grade D at baseline, grade B at 1-month and grade A at 3-months). MRIs at 1-month (F and G) demonstrated a partial response and at 3-months demonstrated a complete response (H and I).*

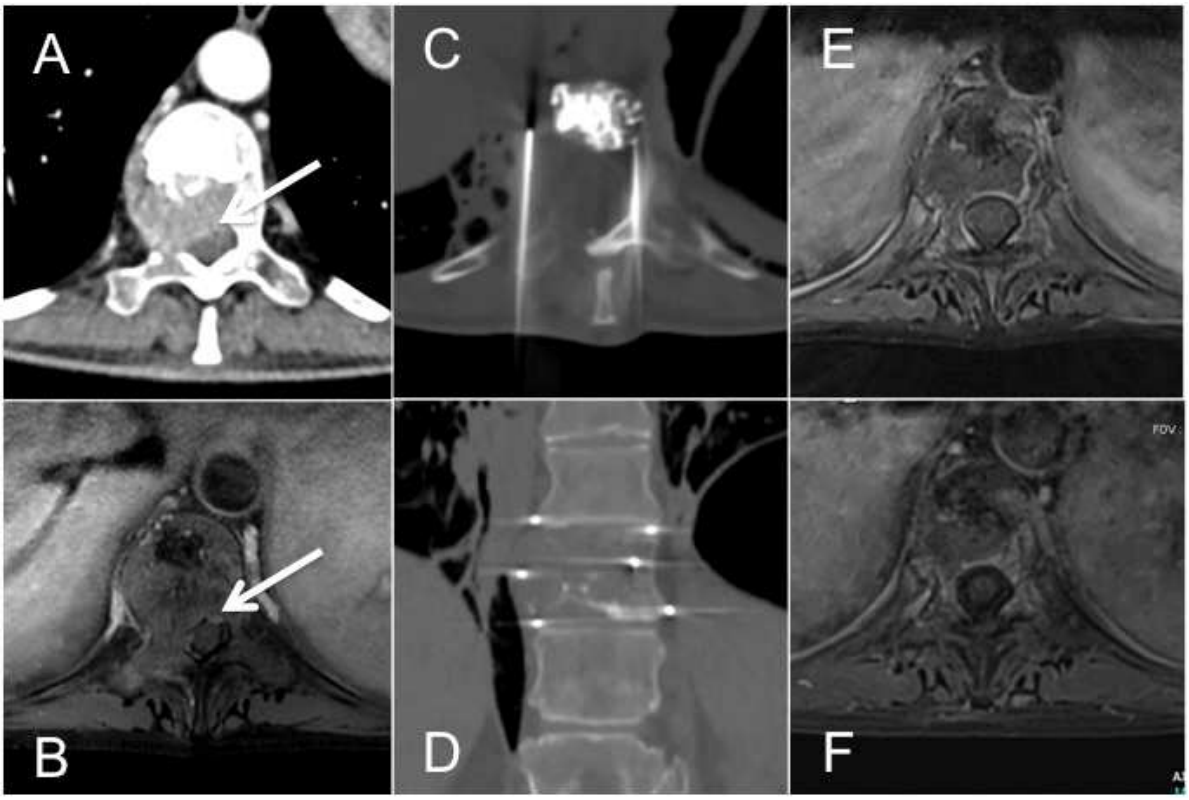


Figure 2 : *A 48-year-old female with a kidney tumor and painful spinal cord compression at T10 level (white arrow). She had received previous radiotherapy (20 Gy in 5 fractions) at the same level*

and cabozantinib was ongoing at the time of analysis. The pretreatment MRI (A) demonstrated a single level epidural spinal cord compression (grade 3) with complete osteolysis of the osseous pedicle and lateral extension in the soft tissues. Electrochemotherapy was performed using 6 electrodes under CT guidance : C axial view (C) and coronal view (D). The patient had marked pain improvement (pain decrease from 5/10 at baseline to 0/10 at 1-month and 3-months) and a marked neurologic benefit (neurological deficit scale: grade B at baseline to grade A at 1-month and 3-months). The MRIs at 1-month (E) and 3-months (F) demonstrated a complete response.

DISCUSSION

MESCC that is radiotherapy-resistant or not candidate for surgical resection frequently leads to severe neurologic symptoms including paraplegia and represents an unmet medical need; electrochemotherapy is a new treatment option in this setting.

ECT significantly improved pain and neurological symptoms in 80% and 55% of our cohort, respectively, with a 77% response rate at 1-month. Complications were overall minor, with the exception of paraplegia in 3 patients who experienced definitive paraplegia as a result of the procedure. Neurological complications can be related to the inevitable technological reality that hyperthermia and irreversible electroporation occur in the immediate vicinity of at the electrodes[19-20]. Further research to tailor this technique is merited to improve the safety profile and risk/benefit balance. It is also important to keep in context that most of the lesions treated would have induced a neurologic deficit such as paraplegia without disease control.

To the best of our knowledge, trans-arterial chemotherapy has been the only minimally invasive strategy that has recently been explored for these patients. Heianna et al. demonstrated that chemoembolization of MESCC refractory to re-irradiation resulted in pain relief in 95% of the 19 patients treated and neurological improvement in 73% of the 15 patients with neurological deficits[21].

Better tumour responses were obtained in hypervascular MESCC, probably because an increased tumour vascularity may increase the total dose of bleomycin delivered to the tumour' cells but also because ECT has

a vascular disrupting action, known as the “vascular lock” effect[22-23], on tumours by destructing small tumour blood vessels, without affecting the larger normal blood vessels surrounding the tumour[24].

Our retrospective single-center study has some limitations. While patient selection criteria were standardized, differences in tumour extent and prior treatment were unavoidable in our small sample size of a wide range of tumour types. Although the operators were technically proficient in CT-guided procedures, we implemented subtle, but nonetheless effective, improvements to our technique given accumulating experience over the 2-year study period. Finally, the follow-up was relatively short, limited to 5.1 months, due to the poor prognosis of this patient population.

CONCLUSIONS

This study provides the first published evidence that ECT can rescue radiotherapy-resistant MESCC, providing rapid and durable pain relief and neurological improvement in the advanced cancer setting.

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